| Semester: VII | | | | | | |
|---|---|---|---|---|---|---|
| **FREIGHT TRANSPORTATION SYSTEMS AND LOGISTICS**<br>**Institutional Elective – II (Group G)**<br>**(Theory)** | | | | | | |
| **Course Code** | **:** | CV375TGH | | **CIE** | **:** | **100 Marks** |
| **Credits: L:T:P** | **:** | 3:0:0 | | **SEE** | **:** | **100 Marks** |
| **Total Hours** | **:** | 42L | | **SEE Duration** | **:** | **3 Hours** |

| Unit-I | 08 Hrs |
|---|---|
| **Characteristics of Freight Transport:** Freight Characteristics, Factors influencing Freight Travel, operators, problems in freight transportation, regional and urban goods travel, intermodal freight travel issues, passenger and freight demand models. | |
| **Unit – II** | **09 Hrs** |
| **Freight Demand Estimation:** Operations, Planning - purpose, process, Data, Freight Agents, costs, Planning Models and Methods-freight demand estimation and forecasting at the regional and urban level, Freight Generation and Freight Trip Generation, Trend and time series models, freight trip rate models, IO model | |
| **Unit –III** | **08 Hrs** |
| **Freight Transport Planning and Operations:** Freight supply – capacity issues; freight productivity and performance; distribution of freight flows; production/consumption to origin/destination, competing modes for specific commodity choice, route planning, scheduling, collection storage, distribution centres, regulation, and enforcement of freight transport. | |
| **Unit –IV** | **09 Hrs** |
| **Logistics and Planning Strategies**: Context of Logistics- Activities of Logistics, Aims of Logistics, Importance of Logistics, Current Trends in Logistics; Logistics Strategy- Strategic Decisions, Logistics Strategy, designing a Logistics Strategy; Locating Facilities- Importance of Location, Choosing the Geographic Region, Infinite Set Approaches, Feasible Set Approaches, Network Models, Location Planning; Planning Resources- Types of Planning, Capacity Planning, Adjusting Capacity, Tactical Planning, Schedules | |
| **Unit –V** | **08 Hrs** |
| **Emerging Trends and Case Studies in Transportation Logistics:** Sustainable and Green Logistics, Digital Transformation in Logistics-IoT, big data analytics, AI, and blockchain in logistics management.<br>**Case Studies and Real-World Applications:** Case studies on logistics and transportation from various sectors. Analysis of successful logistics systems and lessons learned. | |

| Course Outcomes | |
|---|---|
| **CO1** | Explain the characteristics, components, and current trends in freight transportation and logistics |
| **CO2** | Analyze and apply freight demand estimation methods and logistics planning strategies for urban and regional transport systems |
| **CO3** | Evaluate freight transport systems and logistics networks with respect to performance, capacity, sustainability, and regulation |
| **CO4** | Design integrated freight and logistics solutions using data-driven models, technology, and strategic planning approaches |

| **Reference Books** | |
|---|---|
| 1 | M. Ben-Akiva, H. Meersman, and E. V. de Voorde, “Freight Transport Modelling” Bingley, U.K.: Emerald Group Publishing, ISBN-13: 9781781902851,2013. |
| 2 | P. K. Sarkar, V. Maitri, and G. J. Joshi, “Transportation Planning: Principles, Practices and Policies”, 3rd ed. New Delhi, India: PHI Learning, ISBN : 9788195161188, 2024 |
| 3 | L. Tavasszy and G. de Jong, “Modelling Freight Transport”. 1st ed. Amsterdam, Netherlands: Elsevier, ISBN-13: 9780124104006,2013. |
| 4 | M. Al-Azzawi, “Freight and Logistics Transport Modelling and Planning: Mathematical Theories and Practical Applications for the Analysis and Forecasting of Freight Transport Systems”. Saarbrücken, Germany: LAP Lambert Academic Publishing, ISBN-13: 9783848428380,2012 |

| **RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)** | | |
|---|---|---|
| **#** | **COMPONENTS** | **MARKS** |
| 1. | **QUIZZES:** Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. **THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.** | **20** |
| 2. | **TESTS:** Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom’s Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). **TWO tests will be conducted**. Each test will be evaluated for **50 Marks**, adding upto 100 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.** | **40** |
| 3. | **EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20**) ADDING UPTO 40 MARKS**. | **40** |
| | **MAXIMUM MARKS FOR THE CIE (THEORY AND PRACTICE)** | **100** |

| **RUBRIC FOR SEMESTER END EXAMINATION (THEORY)** | | |
|---|---|---|
| **Q. NO.** | **CONTENTS** | **MARKS** |
| | **PART A** | |
| 1 | Objective type questions covering entire syllabus | 20 |
| | **PART B**<br>(Maximum of TWO Sub-divisions only) | |
| 2 | Unit 1 : (Compulsory) | 16 |
| 3 & 4 | Unit 2 : Question 3 or 4 | 16 |
| 5 & 6 | Unit 3 : Question 5 or 6 | 16 |
| 7 & 8 | Unit 4 : Question 7 or 8 | 16 |
| 9 & 10 | Unit 5: Question 9 or 10 | 16 |
| | **TOTAL** | **100** |